12TH CHEMISTRY · CHAPTER 20 · SOLVED EXERCISE

Aromatic Hydrocarbons

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. Multiple Choice Questions:

I. Which of the following is the major product when methyl chloride (CH_3Cl) reacts with methylbenzene in the presence of $AlCl_3$:

(a) 1, 2-dimethyl benzene
(b) 1,4-dimethyl benzene
(c) 1,3-dimethyl benzene
(d) both (a) and (b)

Answer: (d) both (a) and (b) — The methyl group is ortho-para (2,4) directing, so the incoming methyl group enters the 2 and 4 positions.

II. Bromine reacts more readily with phenol than with benzene. This is because the:

(a) benzene ring in phenol is more susceptible to nucleophilic attack
(b) benzene ring in phenol is deactivated because the oxygen of the OH group is very electronegative
(c) lone pair of electrons on the oxygen atom in phenol overlap with the delocalised electrons in the benzene ring
(d) lone pair of electrons on the oxygen atom enable phenol to act as an electrophile

Answer: (c) lone pair of electrons on the oxygen atom in phenol overlap with the delocalised electrons in the benzene ring — The overlap increases the electron density of the ring, which activates it towards electrophiles.

III. Aluminium chloride catalyses the reaction of benzene and ethanoyl chloride. This is because aluminium chloride.

(a) has significant covalent character
(b) exists as a dimer in non-polar solvents
(c) is an electron pair acceptor
(d) is rapidly hydrolysed

Answer: (c) is an electron pair acceptor — As a Lewis acid it accepts a lone pair from the chlorine of CH_3COCl, generating the acylium ion electrophile.

IV. Benzene is nitrated using a mixture of concentrated nitric and sulphuric acids. The sulphuric acid

(a) acts as a solvent for the benzene and the nitric acid
(b) protonates the benzene to speed up the reaction
(c) protonates the nitric acid which acts as a base
(d) reacts with the benzene to form a benzenesulfonic acid intermediate

Answer: (c) protonates the nitric acid which acts as a base — Protonated nitric acid loses water to give the nitronium ion, NO_2^+.

studysapiens.com

V. In the reaction of benzene with chloromethane, aluminium chloride is added because it reacts with

(a) benzene to produce an electrophile
(b) benzene to produce a nucleophile
(c) chloromethane to produce a nucleophile
(d) chloromethane to produce an electrophile

Answer: (d) chloromethane to produce an electrophile — $CH_3Cl + AlCl_3 \rightarrow CH_3^+ + [AlCl_4]^-$; the carbocation is the electrophile.

VI. Methylbenzene reacts with a mixture of concentrated nitric acid and concentrated sulphuric acid. What is the name of the mechanism for this reaction?

(a) electrophilic addition
(b) electrophilic substitution
(c) nucleophilic addition
(d) nucleophilic substitution

Answer: (b) electrophilic substitution — NO_2^+ (electrophile) replaces a hydrogen atom of the ring.

VII. X-ray diffraction provides evidence that benzene molecules have:

(a) delocalised π-electrons
(b) carbon-carbon bonds that are all the same length
(c) lower thermodynamic stability than molecules of 1,3,5-cyclohexatriene
(d) greater thermodynamic stability than molecules of 1,3,5-cyclohexatriene

Answer: (b) carbon-carbon bonds that are all the same length — X-ray studies measure bond lengths: all C–C bonds are 139 pm. Stability is shown by heats of hydrogenation, not by X-rays.

VIII. Which statement(s) support(s) the delocalised model for the structure of benzene? 1 All carbon–carbon bonds have the same length. 2 The enthalpy change of hydrogenation of benzene is less exothermic than expected. 3 Bromine reacts with benzene less readily than with cyclohexene.

(a) 1, 2 and 3
(b) Only 1 and 2
(c) Only 2 and 3
(d) Only 1

Answer: (a) 1, 2 and 3 — Equal bond lengths, the lower heat of hydrogenation (208 instead of 360 kJ/mol) and the low reactivity towards bromine are all explained by delocalisation.

IX. Benzene reacts with an organic reagent in the presence of a halogen carrier to form phenylethanone. Which organic reagent is required?

(a) CH_3CH_2OH
(b) CH_3CHO
(c) CH_3COCl
(d) CH_3COOH

Answer: (c) CH_3COCl — Friedel-Crafts acylation: $C_6H_6 + CH_3COCl \rightarrow C_6H_5COCH_3 + HCl$ ($AlCl_3$).

The exercise has nine MCQs.

Q2. SHORT ANSWER QUESTIONS

a) Benzene reacts with chlorine in the presence of a halogen carrier, such as $AlCl_3$.

i) Write the equation for the reaction of benzene with chlorine.

$$C_6H_6 + Cl_2 \rightarrow C_6H_5Cl + HCl\ (AlCl_3)$$

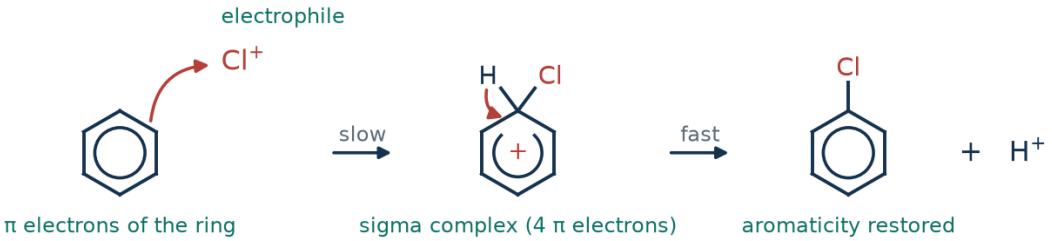
ii) How does the halogen carrier allow the reaction to take place?

$AlCl_3$ is a Lewis acid (electron pair acceptor). It coordinates to a chlorine molecule, polarises the Cl–Cl bond and generates the strong electrophile Cl^+, which benzene can react with.

$$Cl_2 + AlCl_3 \rightarrow Cl^+ + [AlCl_4]^-$$

iii) Outline a mechanism for this reaction. Include curly arrows and relevant dipoles.

$$Cl^{\delta+}\text{–}Cl^{\delta-} + AlCl_3 \rightarrow Cl^+ + [AlCl_4]^-$$



$$H^+ + [AlCl_4]^- \rightarrow HCl + AlCl_3$$

iv) State the name of this mechanism.

Electrophilic substitution (electrophilic aromatic substitution).

b) Diphenylethanedione is a pale yellow colour, which disappears when it is reduced. The colour results from the arrangement of the delocalised π-bond electrons. Explain what is meant by the term delocalised π-bond electrons.

Delocalised π-bond electrons are π electrons that are not fixed between two particular carbon atoms. They are formed by the sideways overlap of parallel p orbitals on several adjacent atoms, so the electrons are spread (shared) over all these atoms, above and below the plane of the molecule, as in the benzene ring where six π electrons are shared equally by six carbon atoms.

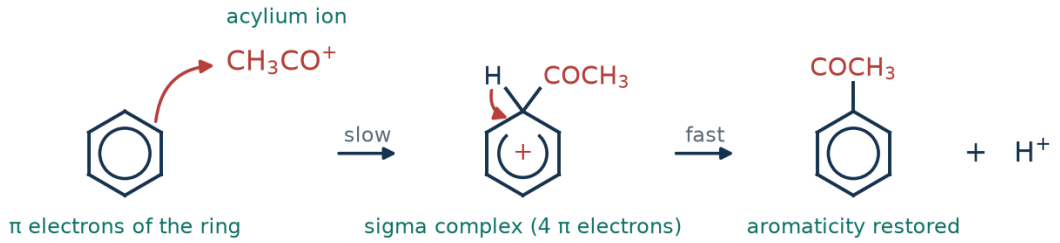
c) An acylium ion has the structure R-C=O where R is any alkyl group. In the conversion of benzene into phenylethanone, $C_6H_5COCH_3$, an acylium ion CH_3CO^+ reacts with a benzene molecule.

i) Write an equation to show the formation of this acylium ion from ethanoyl chloride and one other substance.

$$CH_3COCl + AlCl_3 \rightarrow CH_3CO^+ + [AlCl_4]^-$$

ii) Name and outline the mechanism of the reaction of this acylium ion with benzene.

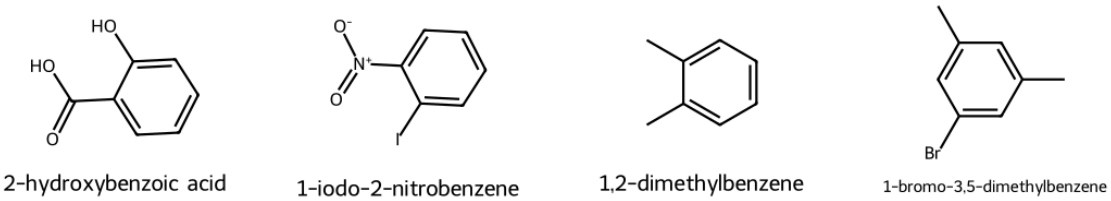
Name: electrophilic substitution (Friedel-Crafts acylation). The π electrons of the ring attack the acylium ion to form a sigma complex, which then loses a proton to restore the aromaticity.



$$H^+ + [AlCl_4]^- \rightarrow HCl + AlCl_3$$

d) Name the following molecules:

From left to right:



(1) 2-hydroxybenzoic acid; (2) 1-iodo-2-nitrobenzene; (3) 1,2-dimethylbenzene; (4) 1-bromo-3,5-dimethylbenzene.

e) Phenol reacts much more readily with bromine than benzene does.

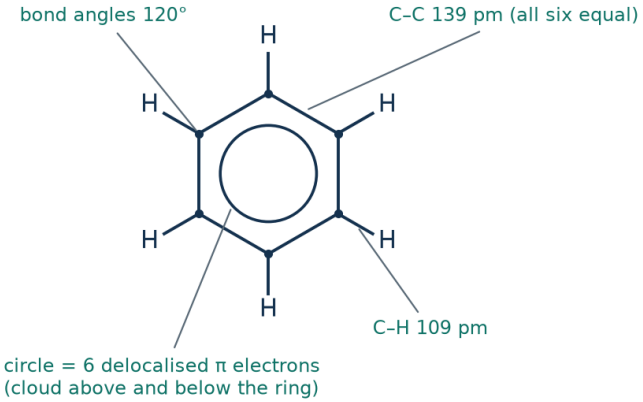
i) Describe, with the aid of a diagram, the bonding in benzene.

σ framework: each carbon atom in benzene is sp^2 hybridized. The three sp^2 hybrid orbitals of each carbon form three σ bonds: two with the adjacent carbon atoms (sp^2–sp^2) and one with a hydrogen atom (sp^2–s). All the sp^2 orbitals lie in the same plane, so all carbon and hydrogen atoms are coplanar with angles of 120°.

π system: the six unhybridized 2p orbitals, one on each carbon, are perpendicular to this σ framework and parallel to each other. They overlap sideways on both sides to form a π-electron cloud above and below the plane of the ring, making it a conjugated

molecule. The six electrons are shared equally among all six carbons; this is called **delocalization** of π electrons.

Delocalization makes all six C–C bond lengths the same (139 pm). So benzene is represented by a hexagon with an inscribed circle, where the circle represents the delocalised π electrons.



Bonding in benzene

ii) Explain why electrophiles, such as bromine, react much more readily with phenol than with benzene.

In phenol, a lone pair of electrons on the oxygen atom of the –OH group overlaps with the delocalised π electrons of the benzene ring. This increases the electron density of the ring (the ring is activated), especially at the 2, 4 and 6 positions, so it attracts and polarises electrophiles such as bromine much more strongly. Benzene has a lower π electron density and needs a Lewis acid (halogen carrier) to generate the electrophile.

Q3. CONSTRUCTED RESPONSE QUESTIONS

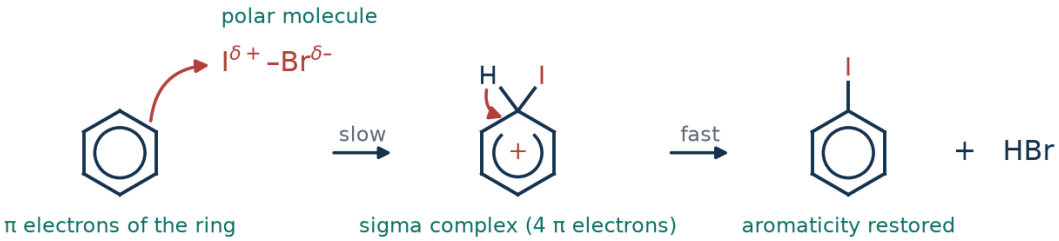
a) The compound iodine monobromide, IBr, also reacts with benzene in an electrophilic reaction.

i) Which compound would be the main product of this reaction, iodobenzene or bromobenzene? Explain your answer.

Iodobenzene. Bromine is more electronegative than iodine, so the I–Br bond is polarised as $I^{\delta+}$–$Br^{\delta-}$. The positive end of the polar molecule, iodine, is the electrophile and bonds to the ring; bromine leaves as Br^-.

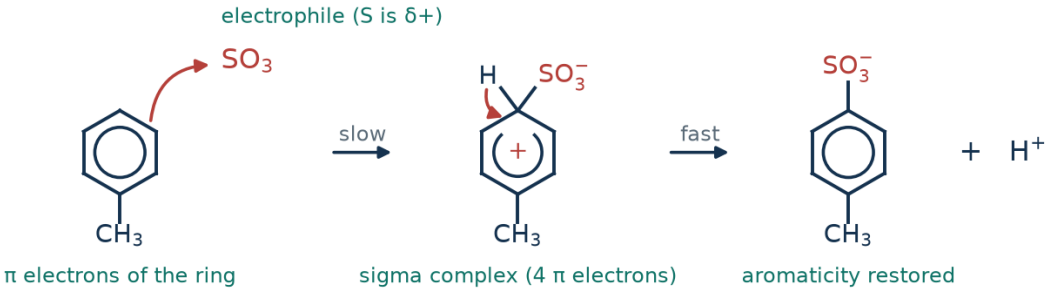
ii) Deduce an equation for the reaction of iodine monobromide with benzene.

$$C_6H_6 + IBr \rightarrow C_6H_5I + HBr$$



b) Methylbenzene reacts with sulphur trioxide, SO_3, to form D, shown below. The electrophile in this reaction is SO_3. Complete the mechanism for the formation of D. Show curly arrows and the structure of the intermediate.

The sulphur atom of SO_3 is δ+ (bonded to three electronegative oxygen atoms). A curly arrow goes from the ring π electrons to S, giving the intermediate (sigma complex) in which the ring carbon para to the CH_3 group carries both H and SO_3^- and the ring has a positive charge. A second curly arrow from the C–H bond back into the ring restores the aromaticity and releases H^+, giving D.



c) This question is about the chemistry of aromatic compounds. Benzoic acid can be nitrated by concentrated nitric acid in the presence of concentrated sulphuric acid as a catalyst, as shown in equation below. The organic product of this reaction is 3-nitrobenzoic acid.

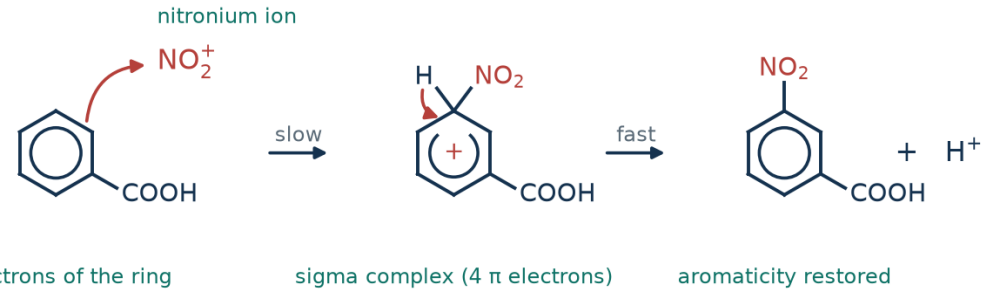
$$C_6H_5COOH + HNO_3 \rightarrow 3\text{-}O_2NC_6H_4COOH + H_2O\ (H_2SO_4)$$

i) Outline the mechanism for this nitration of benzoic acid. Show how H_2SO_4 behaves as a catalyst.

Formation of the electrophile:

$$H_2SO_4 + HNO_3 \rightleftharpoons NO_2^+ + HSO_4^- + H_2O$$

Electrophilic attack and loss of H^+: –COOH is meta directing, so NO_2^+ attacks the 3-position.



Catalyst: the H^+ released combines with HSO_4^- to re-form H_2SO_4, so sulphuric acid is not used up.

$$HSO_4^- + H^+ \rightarrow H_2SO_4$$

ii) A chemist carries out the reaction in Equation above using 4.97 g of benzoic acid. The chemist obtains 3-nitrobenzoic acid as an impure solid. The chemist purifies the solid to obtain 4.85 g of 3-nitrobenzoic acid. Describe a method to obtain a pure sample of 3-nitrobenzoic acid from the impure solid, determine the percentage yield and check its purity.

Purification (recrystallisation):

1. Dissolve the impure solid in the minimum volume of hot solvent (e.g., hot water).
2. Filter the hot solution to remove insoluble impurities.
3. Cool the filtrate so that pure crystals form; soluble impurities stay in solution.
4. Filter the crystals (under reduced pressure), wash with a little cold solvent and dry.

Percentage yield:

$M_r(C_6H_5COOH) = 122$; $M_r(O_2NC_6H_4COOH) = 167$

moles of benzoic acid = 4.97 / 122 = 0.0407 mol

theoretical mass of product = 0.0407 × 167 = 6.80 g

percentage yield = (4.85 / 6.80) × 100 = 71.3 %

Checking purity: measure the melting point. A pure sample melts sharply at the literature value (about 140–142 °C); an impure sample melts lower and over a range.

Recrystallisation and the melting-point check are standard practical methods; they are not described in this chapter. The melting point quoted is from data tables.

DESCRIPTIVE QUESTIONS

Q4. In contrast to benzene, the reaction of an alkene with bromine does not need a halogen carrier. Compare the different reactivities of benzene and alkenes towards chlorine.

	Alkenes	Benzene
π electrons	Localised between two carbon atoms: high electron density.	Delocalised over six carbon atoms: lower electron density between any two carbons.
Effect on the halogen	The π bond polarises the halogen molecule by itself; no catalyst is needed.	Cannot polarise the halogen enough; a halogen carrier ($AlCl_3$, $FeCl_3$) is needed to generate Cl^+.
Reactivity	Very reactive; react readily at room temperature.	Very stable (resonance energy 152 kJ mol^{-1}); less reactive.
Type of reaction	Electrophilic addition.	Electrophilic substitution; addition would destroy the delocalisation.
Product with chlorine	A dichloroalkane, e.g., $CH_2{=}CH_2 + Cl_2 \rightarrow CH_2Cl{-}CH_2Cl$.	Chlorobenzene: $C_6H_6 + Cl_2 \rightarrow C_6H_5Cl + HCl$ ($AlCl_3$).

Benzene adds chlorine only under forcing conditions (UV light and heat), giving 1,2,3,4,5,6-hexachlorocyclohexane.

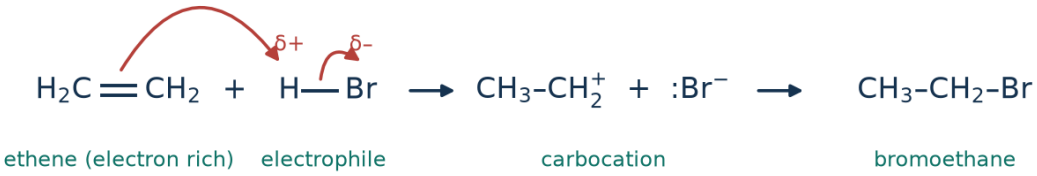
In electrophilic substitution a hydrogen atom of the ring is replaced. The reaction passes through an intermediate in which the aromaticity is broken, but it is restored at the end, so the overall stability of the benzene ring (extensive delocalisation of π electrons) is not affected. In electrophilic addition the delocalisation would be permanently broken, so addition is not favourable.

Q5. Many organic reactions use electrophiles as reagents. Explain the role of electrophiles in organic chemistry. Your answer should include one reaction of an aliphatic compound and one reaction of an aromatic compound, including relevant mechanisms.

An **electrophile** is an atom or group of atoms that is able to accept a pair of electrons to form a new bond. It is a positive ion or the positive end of a polar molecule, e.g., H^+, Br^+, NO_2^+, SO_3. Electrophiles attack electron-rich positions: the π electrons of a C=C double bond or of a benzene ring. A new σ bond is formed with the electrophile, giving a positively charged intermediate.

Aliphatic compound, electrophilic addition: the π bond of ethene attacks the δ+ hydrogen of H–Br, giving a carbocation, which is then attacked by the bromide ion. The electrophile is added and the double bond is lost.

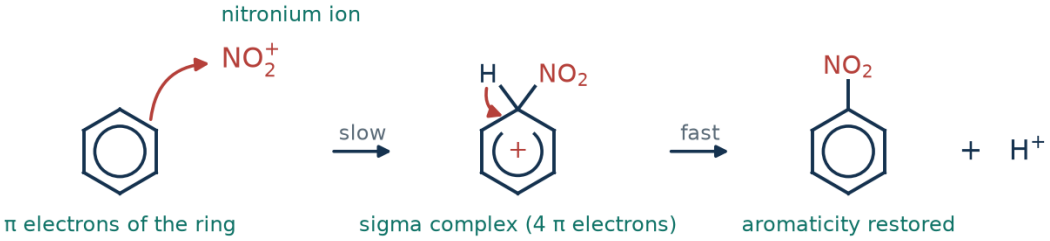
$$CH_2{=}CH_2 + HBr \rightarrow CH_3{-}CH_2{-}Br$$



Aromatic compound, electrophilic substitution: in nitration the electrophile NO_2^+ is generated from conc. HNO_3 and conc. H_2SO_4. The π electrons of the ring attack it to form a sigma complex; loss of H^+ then restores the aromaticity. The electrophile replaces a hydrogen atom and the ring stays intact.

$$H_2SO_4 + HNO_3 \rightleftharpoons NO_2^+ + HSO_4^- + H_2O$$

$$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O \text{ (conc. } H_2SO_4\text{, 50 °C)}$$



So the role of the electrophile is the same in both, but the alkene gives an addition product while benzene gives a substitution product, because of the stability of the delocalised ring.

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 20.1

a) The delocalization (resonance) energy of benzene is about 152 kJ mol^{-1}. Explain what that means and how it affects the reactivity of benzene.

It means that benzene is 152 kJ mol^{-1} more stable (lower in energy) than the hypothetical 1,3,5-cyclohexatriene with three separate double bonds: its heat of hydrogenation is 208 kJ mol^{-1} instead of the expected 360 kJ mol^{-1}. This extra stability comes from the delocalisation of the π electrons. As a result benzene is less reactive than alkenes: it resists addition reactions, which would destroy the delocalisation, and undergoes electrophilic substitution instead.

b) The X-ray studies show that benzene has a planar structure. If benzene were not planar, what difference in the reactivity of benzene would be seen?

If the ring were not planar, the six 2p orbitals would not be parallel and could not overlap sideways all round the ring. The π electrons would not be delocalised, and there would be no resonance stabilisation. Benzene would then have ordinary localised double bonds and would react like an alkene: it would be much more reactive and would undergo addition reactions easily.

c) Why all the bonds in the benzene ring are of equal bond length?

The six π electrons are delocalised and shared equally among all six carbon atoms, so there are no separate single and double bonds. Every C–C bond is identical, with a length of 139 pm, between a single bond (154 pm) and a double bond (133 pm).

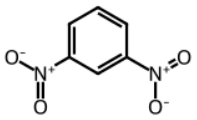
d) How the atomic orbital concept explains the delocalization of π electrons in benzene?

Each carbon is sp^2 hybridized and uses its three hybrid orbitals for σ bonds in one plane. Each carbon is left with one unhybridized 2p orbital, perpendicular to the ring and parallel to the others. Each p orbital overlaps sideways with the p orbitals on both sides, forming a continuous π-electron cloud above and below the plane of the ring. The six electrons are therefore shared equally by all six carbons, i.e., delocalised.

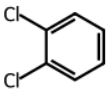
Quick Check 20.2

Give names to following arenes:

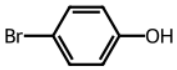
No.	Substituents on the ring	Name
i)	NO_2 at positions 1 and 3	1,3-dinitrobenzene
ii)	Cl at positions 1 and 2	1,2-dichlorobenzene
iii)	OH at 1, Br at 4	4-bromophenol
iv)	CH_3 at 1, Cl at 2 and 4	2,4-dichlorotoluene (2,4-dichloro-1-methylbenzene)
v)	CH_3 at 1, NO_2 at 2, 4 and 6	2,4,6-trinitrotoluene (2-methyl-1,3,5-trinitrobenzene)
vi)	$-COOCH_3$ group	methyl benzoate



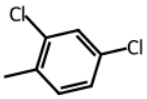
i) 1,3-dinitrobenzene



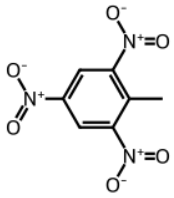
ii) 1,2-dichlorobenzene



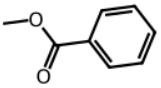
iii) 4-bromophenol



iv) 2,4-dichlorotoluene



v) 2,4,6-trinitrotoluene



vi) methyl benzoate

Quick Check 20.3

Although benzene is highly unsaturated yet it does not prefer to undergo addition reactions. Explain.

The six π electrons of benzene are delocalised over the whole ring, which gives it extra stability (resonance energy 152 kJ mol^{-1}). An addition reaction would break this delocalisation permanently and the stability would be lost, so addition is not favourable. In substitution the aromaticity is broken only in the intermediate and is restored at the end, so benzene prefers electrophilic substitution.

Quick Check 20.4

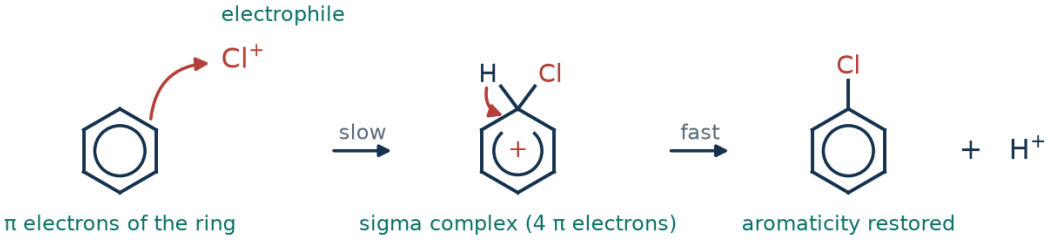
a) Write mechanism of reaction of benzene with chlorine using $AlCl_3$ as catalyst.

$$C_6H_6 + Cl_2 \rightarrow C_6H_5Cl + HCl \text{ (anhydrous } AlCl_3)$$

Step 1, generation of the electrophile: the Lewis acid accepts a lone pair from chlorine and polarises the Cl–Cl bond.

$$Cl^{\delta+}\text{–}Cl^{\delta-} + AlCl_3 \rightarrow Cl^+ + [AlCl_4]^-$$

Step 2, electrophilic attack: the π electrons of benzene attack Cl^+, forming the sigma complex. **Step 3, restoration of aromaticity:** $[AlCl_4]^-$ removes H^+ from the sigma complex.



$$H^+ + [AlCl_4]^- \rightarrow HCl + AlCl_3$$

Quick Check 20.5

Show how the electrophile is produced in the nitration of benzene from sulphuric acid and nitric acid. Which of these two acts as an acid and which acts as a base?

$$H_2SO_4 + HNO_3 \rightleftharpoons NO_2^+ + HSO_4^- + H_2O$$

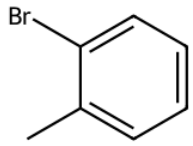
Sulphuric acid is the stronger acid: it donates a proton to nitric acid, so **H_2SO_4 acts as an acid**. Nitric acid accepts the proton, so **HNO_3 acts as a base**. The protonated nitric acid then loses water to give the nitronium ion, NO_2^+, the electrophile.

Quick Check 20.6

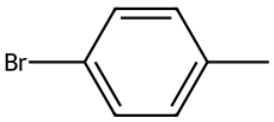
1. **Write down reactions of methyl benzene with bromine in the presence of $AlBr_3$.**

The methyl group is 2,4-directing, so a mixture of 2-bromomethylbenzene and 4-bromomethylbenzene is formed:

$$2C_6H_5CH_3 + 2Br_2 \rightarrow 2\text{-}BrC_6H_4CH_3 + 4\text{-}BrC_6H_4CH_3 + 2HBr\ (AlBr_3)$$



2-bromomethylbenzene



4-bromomethylbenzene

2. **Explain the difference in the reactivity of bromine with benzene and with phenol. Include the type of reaction, the products that form, and any conditions required. Mechanisms for the reactions are not required.**

	Benzene	Phenol
Type of reaction	Electrophilic substitution	Electrophilic substitution
Conditions	Liquid bromine with a Lewis acid (halogen carrier) such as $FeBr_3$ or $AlBr_3$	Bromine water at room temperature; no catalyst needed
Products	Bromobenzene and HBr (one Br enters)	2,4,6-tribromophenol (white precipitate) and 3HBr
Reactivity	Less reactive	Much more reactive: –OH activates the 2, 4 and 6 positions

$$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr$$

$$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$$

In phenol, a lone pair of electrons on the oxygen atom of the –OH group overlaps with the delocalised π electrons of the benzene ring. This increases the electron density of the ring (the ring is activated), especially at the 2, 4 and 6 positions, so it attracts and polarises electrophiles such as bromine much more strongly. Benzene has a lower π electron density and needs a Lewis acid (halogen carrier) to generate the electrophile.

The textbook states that –OH directs incoming groups to the 2, 4 and 6 positions; the bromine-water conditions and the white precipitate of 2,4,6-tribromophenol are from the phenol chapter (Ch. 22).

3. Explain the difference in the reactivity of bromine with benzene and with ethyl benzene. Include the type of reaction, the products that form, and any conditions required. Mechanisms for the reactions are not required.

Both react by electrophilic substitution with bromine and a Lewis acid ($AlBr_3$ or $FeBr_3$) in the absence of UV light. Benzene gives bromobenzene. Ethylbenzene reacts faster, because the ethyl group donates electrons to the ring and activates it; it is 2,4-directing, so a mixture of 1-bromo-2-ethylbenzene and 1-bromo-4-ethylbenzene is formed.

$$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr\ (AlBr_3)$$

$$2C_6H_5C_2H_5 + 2Br_2 \rightarrow 2\text{-}BrC_6H_4C_2H_5 + 4\text{-}BrC_6H_4C_2H_5 + 2HBr\ (AlBr_3)$$

4. Which of 1,1-dimethyethyllbenzene and methyl benzene is easier to form by Friedel-Craft reaction? Explain based on the stability of carbocation.

1,1-Dimethylethylbenzene is easier to form. The electrophile in Friedel-Crafts alkylation is a carbocation. For 1,1-dimethylethylbenzene it is the tertiary carbocation $(CH_3)_3C^+$, which is stabilised by three electron-donating methyl groups and forms easily from $(CH_3)_3CCl$ and $AlCl_3$. For methylbenzene it would be the methyl carbocation, CH_3^+, which has no alkyl groups and is the least stable carbocation, so it is difficult to form.

Quick Check 20.7

a) Explain why phenol can be nitrated using milder conditions than for benzene.

The –OH group is an activating group: a lone pair on its oxygen is donated into the ring, increasing the electron density of the ring. The activated ring is attacked by the electrophile much more easily, so dilute nitric acid at room temperature is enough. Benzene needs conc. HNO_3 with conc. H_2SO_4 at about 50 °C.

b) How would you prepare 3-chloronitrobenzene from benzene?

Nitrate first, then chlorinate. The nitro group is meta directing, so chlorine enters the 3-position.

$$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O\ (\text{conc. } H_2SO_4,\ 50\ °C)$$

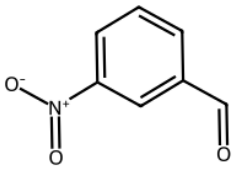
$$C_6H_5NO_2 + Cl_2 \rightarrow 3\text{-}ClC_6H_4NO_2 + HCl\ (AlCl_3)$$

If chlorination were done first, the product would be the 2- and 4-isomers.

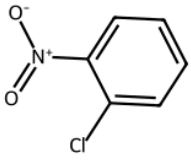
c) Which products are expected to form in majority when the following are nitrated with conc. sulphuric acid and nitic acid? Give reactions. i. benzaldehyde ii. chlorobenzene iii. phenol iv. Benzene carboxylic acid

	Compound	Group	Major product(s)
i.	benzaldehyde	–CHO (C=O), meta directing	3-nitrobenzaldehyde
ii.	chlorobenzene	–Cl, ortho-para directing	2-nitrochlorobenzene and 4-nitrochlorobenzene
iii.	phenol	–OH, ortho-para directing	2-nitrophenol and 4-nitrophenol
iv.	benzenecarboxylic acid	–COOH (C=O), meta directing	3-nitrobenzoic acid

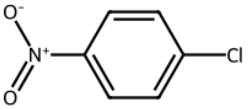
$$C_6H_5CHO + HNO_3 \rightarrow 3\text{-}O_2NC_6H_4CHO + H_2O$$
$$2C_6H_5Cl + 2HNO_3 \rightarrow 2\text{-}O_2NC_6H_4Cl + 4\text{-}O_2NC_6H_4Cl + 2H_2O$$
$$2C_6H_5OH + 2HNO_3 \rightarrow 2\text{-}O_2NC_6H_4OH + 4\text{-}O_2NC_6H_4OH + 2H_2O$$
$$C_6H_5COOH + HNO_3 \rightarrow 3\text{-}O_2NC_6H_4COOH + H_2O$$



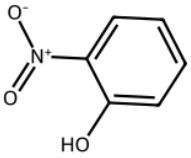
3–nitrobenzaldehyde



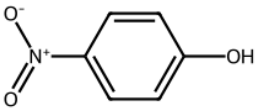
2–nitrochlorobenzene



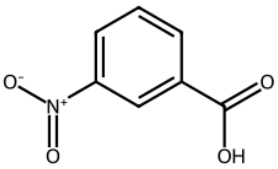
4–nitrochlorobenzene



2–nitrophenol



4–nitrophenol



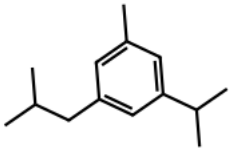
3–nitrobenzoic acid

Halogens are not in the textbook's list of directing groups; chlorine is ortho-para directing (general knowledge).

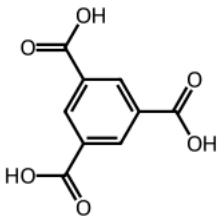
Quick Check 20.8

a) Predict the product formed when: i. the compound given here is oxidized using acidified $KMnO_4$ or $K_2Cr_2O_7$. ii. Bromine is added to chlorobenzene in the presence of $AlBr_3$. iii. Nitrobenzene is reduced in the presence of Ni at high temperature and pressure.

i. The ring carries three alkyl groups at positions 1, 3 and 5, and each has at least one benzylic hydrogen. Each alkyl group is oxidized to –COOH whatever its size, so the product is **benzene-1,3,5-tricarboxylic acid**.



compound given



benzene-1,3,5-tricarboxylic acid

ii. Electrophilic substitution; chlorine is ortho-para directing, so a mixture of **1-bromo-2-chlorobenzene and 1-bromo-4-chlorobenzene** (with HBr) is formed.

iii. With hydrogen and nickel at high temperature and pressure the benzene ring is hydrogenated to a cyclohexane ring; the nitro group is also reduced to $-NH_2$. The product is **cyclohexanamine**, $C_6H_{11}NH_2$ (via phenylamine).

$$C_6H_5NO_2 + 6H_2 \rightarrow C_6H_{11}NH_2 + 2H_2O \text{ (Ni, heat, pressure)}$$

i. The structure drawn has methyl, 1-methylethyl and 2-methylpropyl groups, but its printed label is "3-Ethyl-5-propyltoluene"; either way the product is the same tricarboxylic acid. iii. The textbook describes only hydrogenation of the ring; reduction of the nitro group is general knowledge. If only the ring is considered, the product would be nitrocyclohexane.

studysapiens.com